

• Typecasting:-

Typecasting is the process of converting one data type into another data type in a program.

ex → char a = 65;
cout << a; → prints A.

EX2:- int main {

char c = 590;

cout << c << endl;

return 0;

}

→ Gives Output N

Char can store 1 byte = 8 bits, so total possible values → $2^8 = 256$
But input given → 590, 590 is longer than storage capacity of character. So,

$$590 \% 256 \Rightarrow 78$$

ASCII value of 78 ⇒ N

So, N is printed as output.

590 was given as input but 78 is stored in binary form.

• CONDITIONAL STATEMENTS:-

Q) Find greatest among two numbers

```
#include <iostream>
```

```
using namespace std;
```

```
int main () {
```

```
int a, b;
```

```
cin >> a >> b;
```

```
if (a > b)
```

```
cout << a;
```

```
else if (b > a)
```

```
cout << b;
```

```
else
```

```
cout << "Equal" << endl;
```

```
return 0;
```

```
}
```

Q> Check given number is even or odd.

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cin >> num;
    if (num % 2 == 0)
        cout << "Even" << endl;
    else
        cout << "Odd" << endl;
    return 0;
}
```

Q> Check given number is Positive, Negative or Zero.

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cin >> num;
    if (num > 0)
        cout << "Positive" << endl;
    else if (num < 0)
        cout << "Negative" << endl;
    else
        cout << "Zero" << endl;
    return 0;
}
```

Q> Find Maximum among three numbers.

```
#include <iostream>
using namespace std;
int main() {
    int a, b, c;
    cin >> a >> b >> c;
    if (a > b && a > c)
        cout << a;
```

else if (b > a && b > c)

cout << b;

else if (c > a && c > b)

cout << c;

else cout << "Equal" << endl;

return 0;

}

Q> WAP to check if given character is Vowel or constant.

#include <iostream>

using namespace std;

int main() {

char c;

cin >> c;

if (C == 'a' || C == 'e' || C == 'i' || C == 'o' || C == 'u' ||
C == 'A' || C == 'E' || C == 'I' || C == 'O' || C == 'U')

cout << "Vowel" << endl;

else

cout << "Consonant" << endl;

return 0;

}

Q> WAP to check if a year is leap year or not.

#include <iostream>

using namespace std;

int main() {

int year;

cin >> year;

if (Year % 400 == 0)

cout << "Leap Year" << endl;

else if (Year % 4 == 0 && Year % 100 != 0)

LOGIC	
IF	Year % 400 == 0
	Ya
	Year % 4 == 0 & Year % 100 != 0
	Tab leap Year hai
	or else nhi hai

```
cout << "Leap Year" << endl;
```

```
else  
    cout << "Not Lap Year" << endl;
```

```
return 0;
```

↳

Q) WAP to check if a given number is divisible by 3 & 5.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int num;
```

```
    cin >> num;
```

```
    if (num % 3 == 0 && num % 5 == 0)
```

```
        cout << "Divisible" << endl;
```

```
    else
```

```
        cout << "Not Divisible" << endl;
```

```
    return 0;
```

↳

• LOOPS: -

Q) WAP to print numbers from 1 to n.

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int n;
```

```
    cin >> n;
```

```
    for (int i = 1; i <= n; i++) {
```

```
        cout << i << endl;
```

```
    }
```

```
    return 0;
```

↳

Q) WAP to print all even numbers between 1 to n.